

DYES AND PAINTS



DYES AND PAINTS are substances that are used to stain or give colour to a range of objects, from textiles and paper to buildings and machinery. The substances that give colour to dyes are called **dyestuffs**, which, when dissolved in water, penetrate the fibres of fabrics by means of a chemical reaction. Pigments form the colour in paints. These are held in place using a varnish-like substance called a **vehicle**, or **binder**, which also binds the pigment to the surface being painted. Throughout history, people have created colour, first by means of natural dyes and pigments, and today by using synthetic ones.

Dyes

Some natural dyes still exist, but most used today are synthetic. These are organic chemicals produced by processing petroleum and coal-tar chemicals such as benzene. Most dyes are used in the textile industry, but are also used in the leather, paper, food, and cosmetics industries. The dyes can be applied to the fibre or fabric using either a direct or indirect process.



Wool can be dyed using a mordant dye, but this dye is now avoided in Western countries due to its use of potentially harmful chemicals.

Indirect dyeing

In some dyeing processes, a number of steps are needed to dye the fibre. In one process, a chemical called a **mordant** is first added to the fibre, which is then dyed. The mordant molecules fix the dye to the fabric.

Fabrics can be coloured using a range of dyes



Direct dyeing

In most industrial dyeing processes today, dyes can enter the fibre and colour it in one step, without the need of a mordant. The dye is dissolved in hot water, strained, and then added to the fabric. Sometimes the dye is mixed with salt to help fix the colour.

Early pigments

The first materials used as pigments were probably coloured clays, which were mixed with water or animal oils to make paint. Dyes made from plants and animals were later used to colour textiles. Common plant dyes included woad, madder, saffron, and turmeric. Animal sources included cochineal (beetle) and the Murex sea snail.



The T-shirt on the left shows how the dye has faded.

This T-shirt shows how the colour has remained fast.

Colour fastness

Two of the most important properties demanded of a dye by clothes manufacturers are its abilities to resist being washed out, and not to fade in the light. The colour fastness of a fabric also varies according to the dyeing process that is used and the type of material that is being dyed.

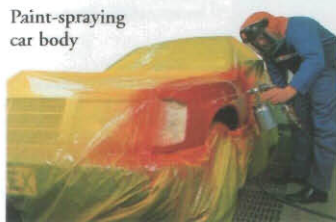
Paints

Paint comes in many colours and can be used as a coating on rigid structures such as houses, bridges, ships, and cars. Finer paints are used by artists to produce imaginative and colourful works of art. The pigments used to make the paints may be natural, such as rutile (titanium dioxide) or synthetic, such as phthalocyanine blue.

Industrial paints

Industrial paints are custom-made for specific jobs. Some paints contain powdered metal and metal oxides, so that the paint can protect exposed structures, such as iron bridges. Paints such as those used on cars are designed to withstand rusting and high temperatures.

Paint-spraying car body



Domestic paints

Most decorating paints are made for easy application. Non-drip paints are jelly-like in the can, but flow easily when applied. Emulsion paint uses water as its vehicle, so splashes can be removed and brushes easily cleaned.

Can of non-drip paint and brush



Oil paints usually come in tubes so that users can squeeze out the exact amount of paint needed.

Oil paints



Artists' paints

Artists use a variety of types of paint to achieve different effects, including watercolours, oils, and acrylics. The pigments in watercolour paints are mixed with a water solution of gum arabic, in oils they are mixed in a slow-drying oil, such as linseed oil, while in acrylics the pigments are mixed with a synthetic resin vehicle.

William Henry Perkin

British chemist William Henry Perkin (1838–1907) accidentally produced the first synthetic dye, mauve, in 1856. He was attempting to make the drug quinine from coal-tar chemicals, but instead produced a purple liquid dye. This was the start of the synthetic dye industry.



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